

Teamwork Agreement

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: Cool Team (Team 9)

Instructor: Kevin Graham

Program/Class: Engineering Pathway Program (EPP)

Members and roles

1. Abdullah Siddiqui - Documentation and Visualization Lead
2. Santiago Martinez - Project Manager
3. Miguel Gonzalez - Quality Assurance & Reproducibility Lead
4. Randal Peterson - Modeling Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord (e.g. Slack)
- Secondary channel: Email (e.g. E-mail for summaries/decisions)
- Response time: 24 hour on weekdays and 48 hour on weekends
- Meeting cadence: 60 minutes, Monday, Thursday
- Decision records: PM logs major decisions in a running document

4) Collaboration & Tools

- Version control: GitHub Repo (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in

README.md

File naming & style: Snake_case, descriptive names

- Backups: e.g. PM tags old versions according to numbered system

5) Quality Standards (Definition of 'Done')

Final deliverables

- 2 unique drone arm sketches with different geometric features
- Thrust-to-weight analysis results clearly displayed, with the best option highlighted
- Results of FEA
- Final design recommendation

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Problem framing, literature/background review, dataset/parameters
- Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.
- Week 3: Validation and quality assurance
- Week 4: Report, slides, presentation rehearsal, repository cleanup

7) Decision-Making & Conflict Resolution

- Default: consensus after discussion; PM facilitates
- If blocked >48 hrs: Time-limited debate, majority vote, PM documents decision
- Technical disagreements: Evidence-based comparison before vote
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md
- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.
- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: Abdullah Siddiqui, ABDULLAH SIDDIQUI Date: 07/15/2026

Name & Signature: MIGUEL GONZALEZ Date: 7/15/2026

Name & Signature: SANTIAGO MARTINEZ Date: 07/15/2026

Name & Signature: RANDAL PETERSON Date: 07/15/2026